

Assignment I: Introduction to Spreadsheet

1. What is a spreadsheet?

A spreadsheet is software used to organize, calculate, and analyze data in rows and columns.

2. How to create a new spreadsheet in MS Excel?

Open Excel → File → New → Blank Workbook.

3. How to create a marksheet using sorting?

Enter student data, apply formulas for total/percentage/grade, then use Sort.

4. How to format a table?

Apply bold headings, center alignment, font size, borders, and heading color.

Assignment II: Advanced Excel Functions

1. What is VLOOKUP?

Searches vertically in first column and returns value from another column.

2. What is HLOOKUP?

Searches horizontally in first row and returns value from specified row.

3. What is XLOOKUP?

Modern lookup function for exact/flexible horizontal or vertical search.

4. What is INDEX()?

Returns value at a specific row and column from a range.

5. What is INDIRECT()?

Converts text into a valid cell reference.

6. What is ADDRESS()?

Returns the cell address as text.

7. What is IF()?

Returns one value if condition is true, another if false.

8. **What is AND()?**

Returns TRUE only if all conditions are true.

9. **What is OR()?**

Returns TRUE if at least one condition is true.

10. **What is NOT()?**

Reverses TRUE to FALSE and FALSE to TRUE.

11. **What is IFS()?**

Evaluates multiple conditions and returns first true result.

12. **What is IFERROR()?**

Returns custom value when formula gives an error.

13. **What is CONCAT()?**

Joins two or more text strings.

14. **What is LEN()?**

Counts number of characters in text.

15. **What is LEFT()?**

Extracts characters from left side of text.

16. **What is RIGHT()?**

Extracts characters from right side of text.

17. **What is MID()?**

Extracts characters from middle of text.

18. **What is TRIM()?**

Removes extra spaces from text.

19. **What is TODAY()?**

Returns current date.

20. What is NOW()?

Returns current date and time.

21. What is EOMONTH()?

Returns last day of a month.

22. What is DATEDIF()?

Calculates difference between two dates.

23. What is NETWORKDAYS()?

Calculates working days between two dates.

24. What is TIME()?

Creates time using hours, minutes, and seconds.

Assignment III: Advanced Excel Concepts

1. What is an array formula?

A formula that performs multiple calculations on one or more items in an array.

2. How to use CSE formula?

Type formula and press Ctrl + Shift + Enter.

3. What is FILTER()?

Returns filtered data based on condition.

4. What is SEQUENCE()?

Generates sequential numbers automatically.

5. What is SORT + UNIQUE?

Extracts unique values and sorts them.

6. What is LARGE()?

Returns nth largest value from a range.

Assignment IV: Advanced Chart Types

1. **What is Histogram?**

Shows frequency distribution of data.

2. **What is Waterfall Chart?**

Shows cumulative positive and negative changes.

3. **What is Gantt Chart?**

Displays project tasks with timeline.

4. **What is Combo Chart?**

Combines two chart types in one graph.

5. **What is Pivot Table?**

Summarizes and analyzes large datasets quickly.

Assignment V: VBA Macro with Loops

1. **What is VBA?**

Visual Basic for Applications used to automate Excel tasks.

2. **What is a Macro?**

A set of VBA instructions to automate repetitive tasks.

3. **What is IF statement in VBA?**

Executes code based on a condition.

4. **What is Select Case?**

Checks one variable against multiple conditions.

5. **What is For...Next loop?**

Repeats code fixed number of times.

6. **What is Do While loop?**

Repeats while condition remains true.

Assignment VI: VBA Macro with Mail

1. How to send mail using VBA?

Use Outlook object in VBA and call `.Send`.

2. What is required?

Installed and logged-in Microsoft Microsoft Outlook